Ball Progression and Player Productivity Across European Leagues

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Hypothesis

Modern football is a vast industry. The global football market is valued at USD 3.43 billion and expected to grow to USD 4.66 billion by 2032 (SkyQuest Technology Consulting Pvt. Ltd., 2024).

Every club in such a large market tries to stand out both through media activity and through victories, so coaches, players, scouts, clubs and even fans often analyze various statistical data related to player and team performance in general. In the pursuit of optimizing player selection, positioning and achieving victories, clubs try to identify which data and factors are the most important and have the greatest impact on results.

The main indicator that brings success is productivity G+A which stands for goals and assists. At first glance this may seem simple, but a huge number of factors influence it such as age, minutes played, position, emotional state, sleep, nutrition and even the weather.

Since there is much more behind these numbers, without going too deep or trying to account for an infinite number of variables, we will focus on one very important and technical factor that illustrates how a football player affects the game. This indicator is how he moves the ball forward. Here there is a distinction between precise, well-thought-out passes (prg_p) and individual ball progressions (prg_r), which we will examine.

It is important to take into account that football differs greatly between leagues in rhythm, style and structure. For example, the English Premier League is known for its speed, physical pressure and high intensity. The Italian Serie A focuses on tactics and defensive organization. In the Spanish La Liga, more attention is given to technique and ball control, where short passing and combination play are common. The Bundesliga emphasizes quick transitions and vertical attacks, while Dutch football is associated with “total football” where players freely change positions and the team plays boldly and openly (LTT Sports & World Football Summit, 2024).

It can be concluded that the same actions can have different effects in different leagues. For example, a progressive run with the ball may be especially effective in a fast-paced and physically demanding league, while in calmer possession-oriented competitions passing and team combinations tend to be more effective. Everything depends on context.

Therefore, in our study we want to test the following hypothesis. In leagues with higher defensive intensity, progressive runs (prg_r) better explain productivity (G+A) than progressive passes (prg_p). To explore this hypothesis, we will use player data, calculate a defensive intensity indicator based on the number of tackles and interceptions, and build an OLS

regression model with interaction variables. This will help us understand how ball progression affects productivity under different tactical conditions.

Data Commentary.

The chosen dataset for analysis was sourced from Kaggle, which collected its data from FBref, a source for football stats including player, team, and league stats. This dataset represents player statistics from the 2025-2026 football season from the top 5 European Leagues. Initially, the dataset contained observations for 2100+ players.

Variable	Explanation
Goals and assists (goals_and_assists)	Dependent numeric variable. This variable shows the total number of goals scored and assists by a player. It serves as a direct indicator of the effectiveness of a player's attacking action.
Progressive Passes (prg_p)	Independent numeric variable. The number of passes that a player makes in order to significantly move the ball forward to the opponent's goal is calculated. This indicator measures a player's contribution to strengthening the team and creating chances by bypassing the opponent's defensive lines with forward passes, improving the data on progressive runs.
Progressive Runs (prg_r)	Independent numeric variable. It counts the number of times a player carries the ball by making a "progressive throw", a throw in which the ball moves a significant distance towards the opponent's goal. This indicator helps to quantify a player's ability to advance the ball by dribbling (maneuvering a ball by one player while moving) and move forward, overcoming defensive lines.
Minutes Played (min)	Independent numeric variable. It reflects the total number of minutes spent by a player on the field during the season. It is used as a control variable to normalize performance indicators and ensure that a player's performance (for example, goals scored) does not depend simply on the fact that he had more opportunities to play.
Age (age)	Independent numeric variable. It reflects the age of the player in years. This factor is used to analyze the relationship between a player's career stage and performance, taking into account potential

	peaks and opportunities related to physical preparation and experience.
Position (pos)	Independent categorical variable. Indicates the main position of the player on the field. For example, Forward (FW), Midfielder (MF), and Defender (DF). This variable is crucial to review performance indicators, as expectations regarding contributions to goals and other actions vary significantly depending on the role.
League (comp)	Independent categorical variable. Defines the national league in which the players' club mainly plays (for example, the Premier League, La Liga). This variable is used to account for internal differences in style, tempo, and quality of play in different leagues, often through clustering or fixed effects in a regression model.

Table 1: Data Commentary of dependent and independent variables

Descriptive Statistics

The frequency distribution analysis performed for the independent categorical variable "Position" which shows that the categories are not equal in frequency and have different proportions, as shown in the table below:

Position	Frequency	Percentage
Forward	518	23.81%
Midfielder	752	34.56%
Defender	778	35.75%
Goalkeeper	128	5.88%
Total	2 176	100%

Table 2: Frequency distribution for the independent variable "Position"

Since football is a team sport with special positional requirements, we expect approximately proportional representation. Usually, a team consists of 3-4 defenders, 3-5 midfielders and 1-3 forwards, and 1 goalkeeper. The observed proportions are largely in line with expectations, although defenders (35.75%) and midfielders (34.56%) slightly outnumber forwards (23.81%). Most notably, goalkeepers (5.88%) are significantly underrepresented compared to the expected ~9.09% (1/11 players), which may reflect that the dataset is unreliable or excludes specialized goalkeeping metrics.

The table of the frequency distribution of the independent categorical, produced for the variable "League", looks like this:

League	Frequency	Percentage
Premier League	432	19.85%
Serie A	460	21.15%
La Liga	475	21.83%
Ligue 1	422	19.39%
Bundesliga	387	17.78%
Total	2 176	100%

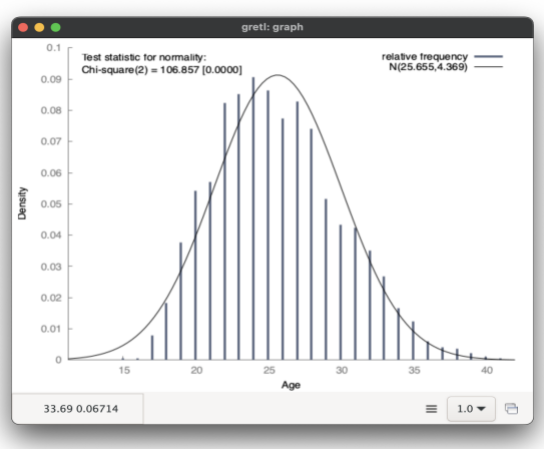
Table 3: Frequency distribution for the independent variable "League"

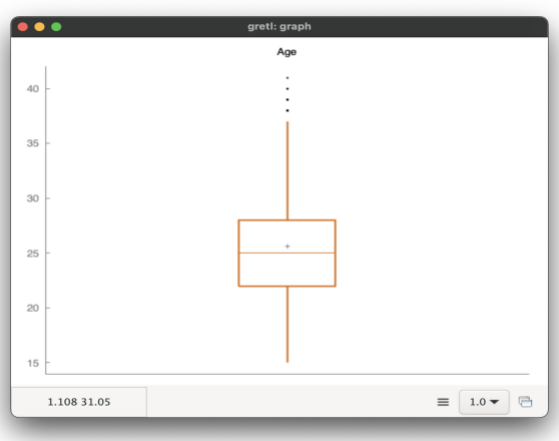
An analysis of the frequency distribution shows that the five major European leagues are relatively balanced. The fact that the segment sizes in the categories range from 17.78% to 21.83% indicates a fairly even distribution across the competitions. This balanced representation minimizes potential bias towards specific leagues, although the interpretation of defensive intensity coefficients should take into account the slight overrepresentation of La Liga (21.83%) and the underrepresentation of the Bundesliga (17.78%).

Variable	Mean	Median	Min	Max	Range	Skewness
Age	25.66	25	15	41	26	0.42
Minutes Played	296.64	283	1	720	719	0.22
Goals and assists	0.68	0	0	14	14	3.11
Progressive Passes	10.88	7	0	98	98	1.75
Progressive Runs	10.78	6	0	115	115	1.93

Table 4: Summary statistics of “Age”, “Minutes Played”, “Goals and assists”, “Progressive Passes”, “Progressive Runs” independent variables

The age variable ranges from 15 to 41 years old, and the median is 25 years old. From a boxplot, the middle 50% of the data between the first and the third quartiles has a range of 6: 25th percentile - 22 years, 75th percentile - 28 years. As expected, most of the players are in the prime of their athletic abilities, with an average age of 25.66 years. The difference between the mean (25.66) and the median (25) of 0.66, combined with a positive skewness coefficient of 0.42, indicates a slight rightward shift in the data, indicating that some veteran players are moving beyond the typical peak of their careers.





Figures 1 and 2: Frequency distribution and Boxplot of “Age” variable, respectively

In the dataset, the number of minutes played ranges from 1 to 720. The relatively small difference between the mean (296.64) and the median (283) of 13.64, along with a low skewness of 0.22, indicates a symmetrical distribution of playing time. This suggests that the dataset mainly includes regular players, rather than including starting line-up players or reserves.

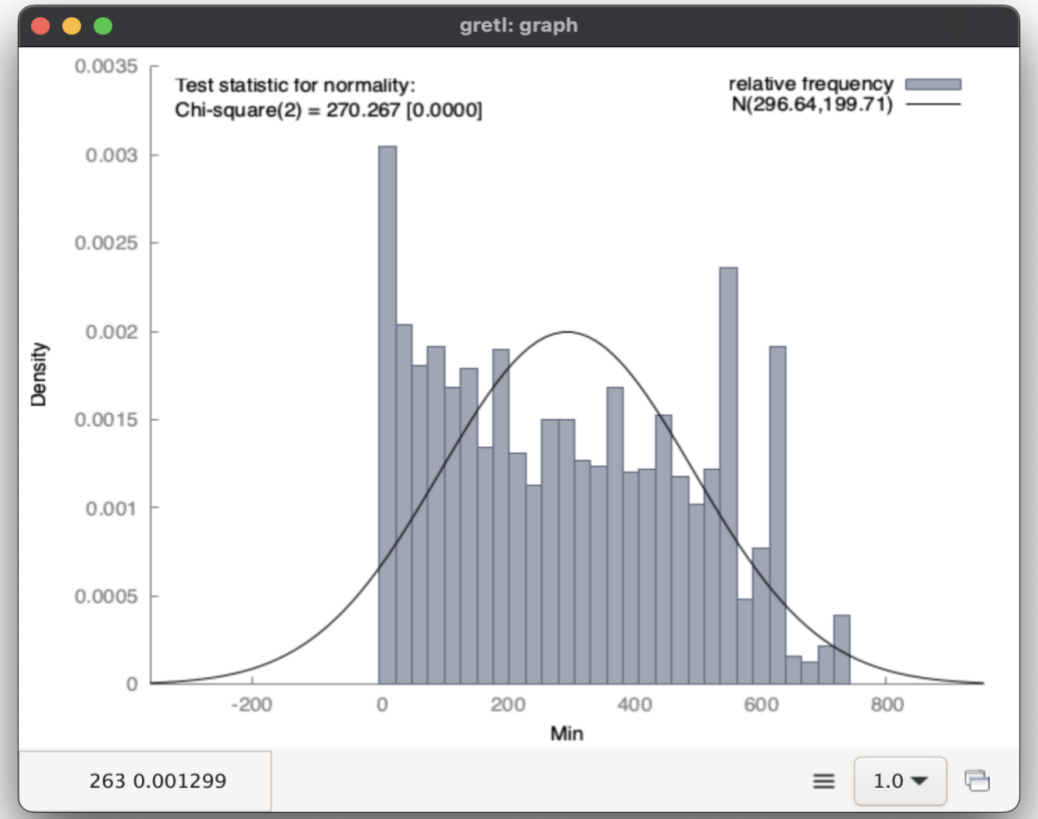


Figure 3: Frequency distribution of “Min” variable

The percentage of goals scored and assists are strongly shifted to the right with a skewness of 3.11, which is typical for performance indicators when most players have limited returns (median = 0), while several top players achieve high results (max = 14). The difference between the mean (0.68) and the median (0) confirms this pattern, indicating that most of the goals scored are scored by a small group of players.

Both progressive passes and progressive runs are skewed to the right, with high values of 1.75 and 1.93, respectively. Meanwhile, their medians are at about half of the means. The pattern suggests that though most players make moderate progressive actions, a subset of key playmakers and dribblers account for a disproportionately high level of these indicators.

Dependent Variable Analysis

Doornik-Hansen, Shapiro-Wilk W, Lilliefors, and Jarque-Bera tests were conducted for the dependent variable “goals_and_assists”, as well as visual diagnostics including a Q-Q plot and kernel density estimation (see Figure 4 and 6). The results from all statistical tests except Shapiro-Wilk W (with p-value 6.24513e-57) yield p-values of 0 or approaching 0, consistently indicating a strong violation of normality. The Q-Q plot (Figure 4) showed a heavy deviation from the diagonal reference line, especially in the upper quantiles, and the kernel density plot (Figure 5) revealed a distribution highly concentrated near zero with a long right tail, confirming strong positive skewness.

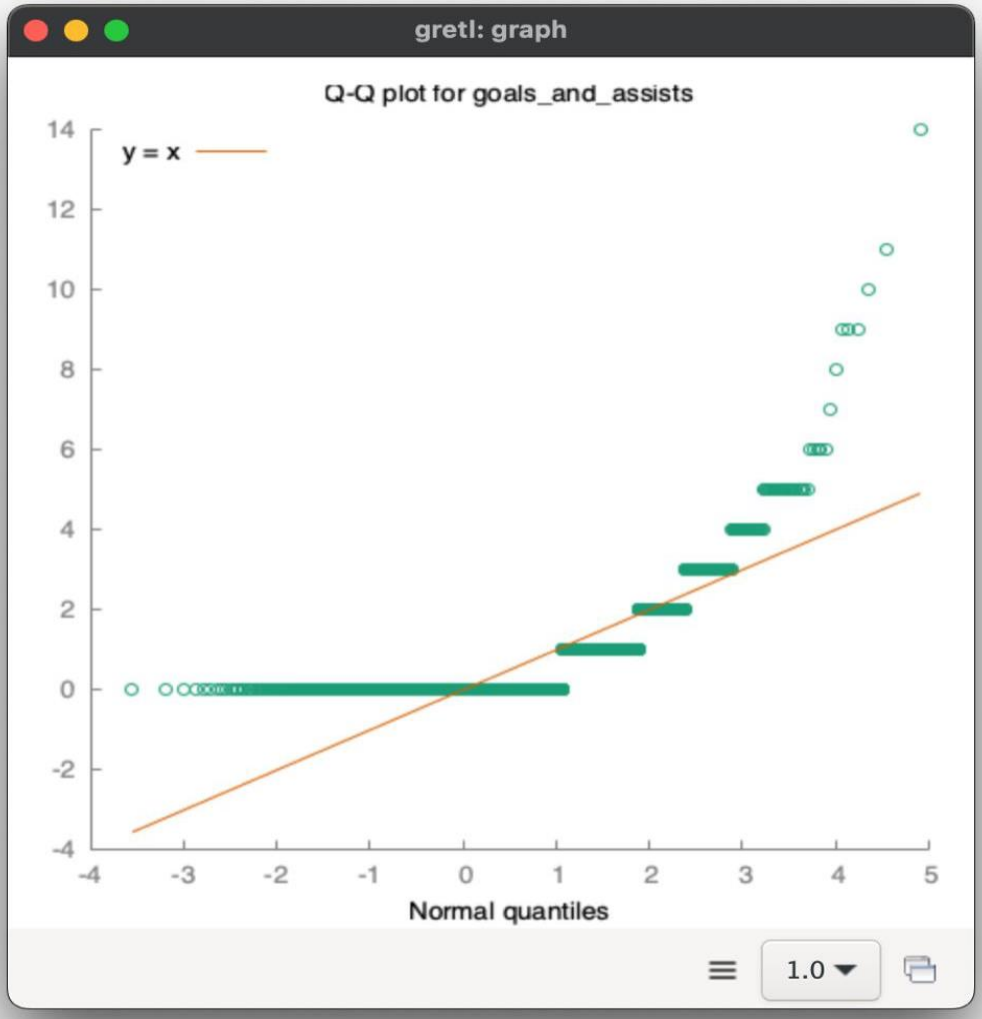


Figure 4: Q-Q plot for “Goals and Assists” variable

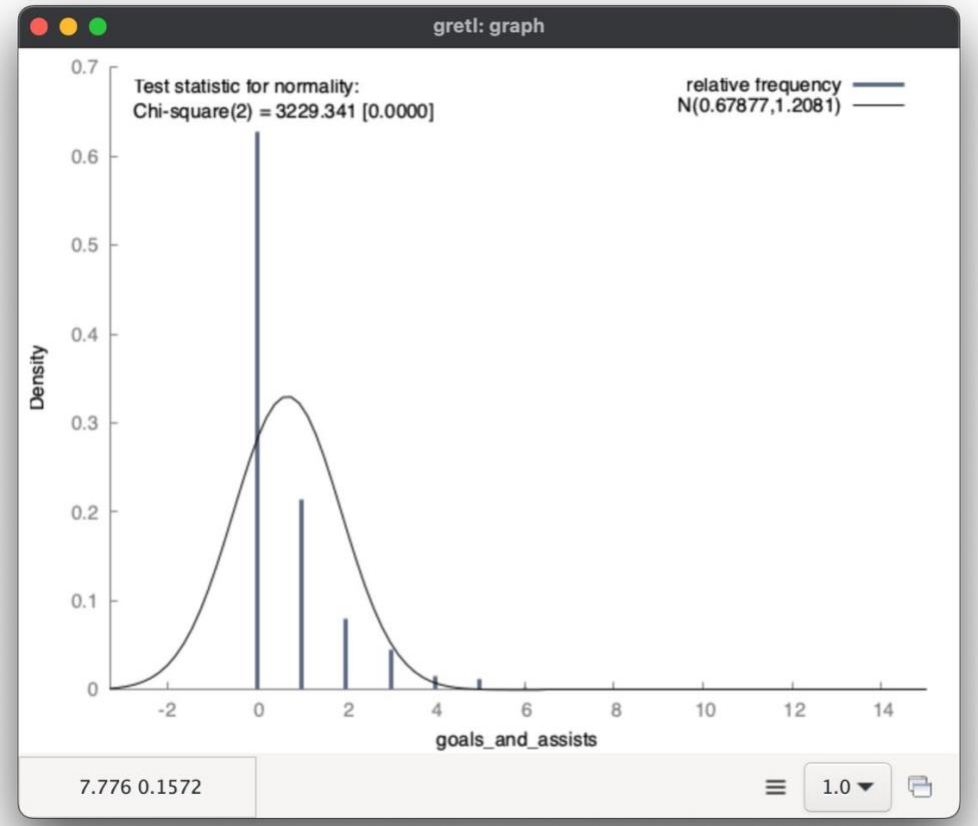


Figure 5: Frequency distribution of “Goals and Assists” variable

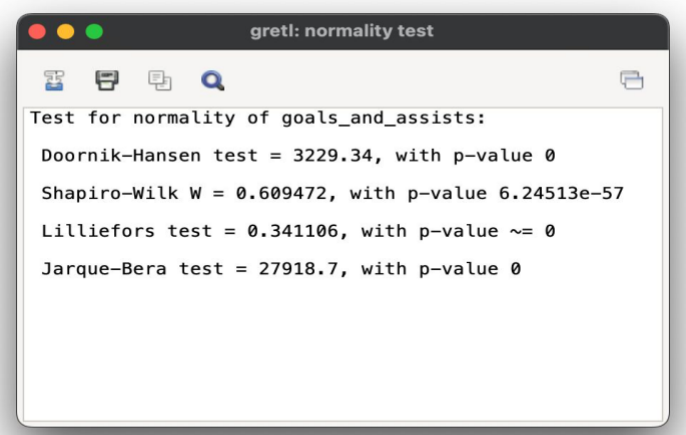


Figure 6: Normality test for “Goals and Assists” variable

The Q-Q plot showed an extreme deviation in the upper quantiles, while the kernel density plot exhibited a strong positive skew, with many observations clustered at zero and a long right tail- typical for count-based performance metrics. To address this, we applied a logarithmic transformation: l_g_and_a_new = log(goals_and_assists + 1).

However, as shown in the updated normality test results (Figure 7) and Q-Q plot (Figure 8), the transformation did not sufficiently normalize the data.

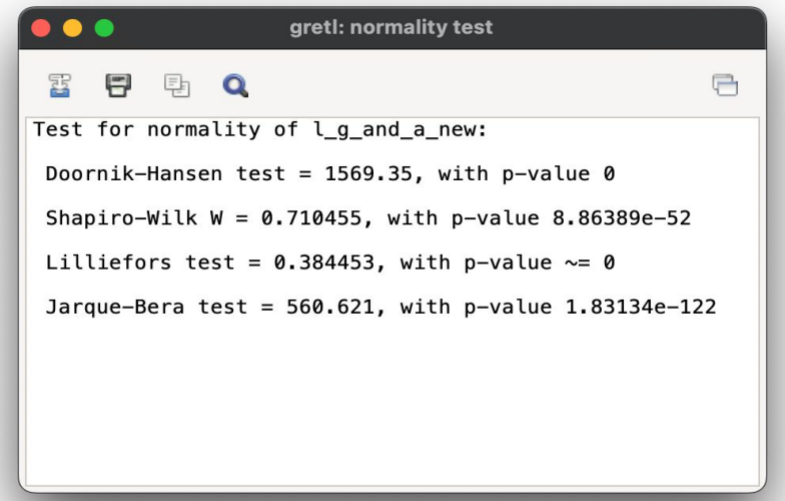


Figure 7: Normality test for a new, logged “Goals and Assists” variable

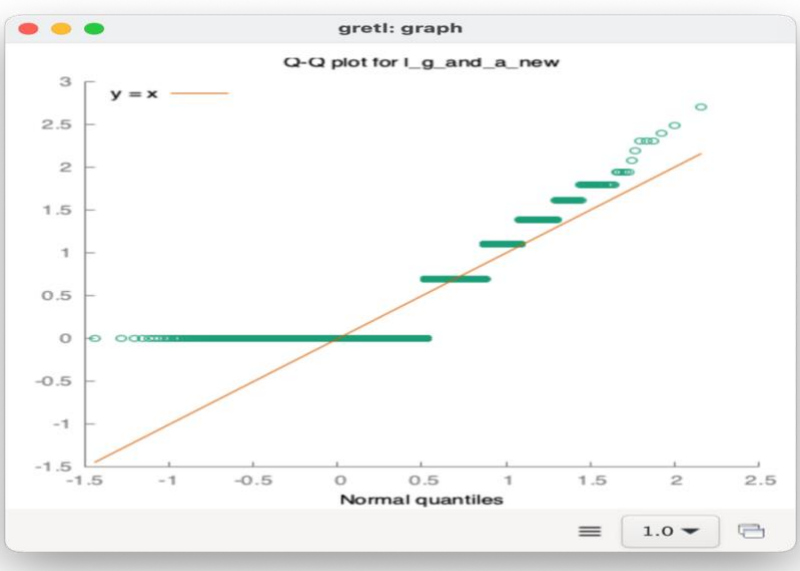


Figure 8: Q-Q plot for a new, logged “Goals and Assists” variable

The raw dependent variable “goals_and_assists” was highly non-normal because of its count-based, right-skewed nature. Applying the transformation log improved the shape of the distribution and brought residuals closer to normality. However, the transformation did not fully normalize the variable. Despite that, the transformed model offers more reliable inference and is suitable for OLS under approximate normality of residuals.

Regression Model Setup

To evaluate the determinants of player offensive productivity, we estimated a log-linear OLS regression model. The dependent variable is (G+A +1) which was chosen to correct for skewness in the original count variable and allow for percentage-based interpretation of coefficients.

The model includes both continuous variables (such as age, minutes played, progressive passes, and progressive runs) and categorical variables for position and league, represented using dummy variables.

We used dummy variables to account for the categorical nature of both player position and league affiliation, as these variables are qualitative and do not have a natural numerical ordering.

A total of seven dummy variables were included:

Three for position:

- $DPos_1$ = Forward
- $DPos_3$ = Defender
- $DPos_4$ = Goalkeeper

Note: Midfielders were excluded and serve as the baseline category.

Four for league:

- $DComp_1$ = Premier League
- $DComp_2$ = La Liga
- $DComp_3$ = Ligue 1
- $DComp_4$ = Serie A

Note: Bundesliga was excluded and serves as the baseline category.

By excluding one category from each set, we avoid the dummy variable trap (perfect multicollinearity) and ensure that coefficients are interpreted relative to the omitted baseline.

This setup allows the model to estimate how each league and position affects a player's productivity relative to the baseline group i.e., midfielders in the Bundesliga, while controlling for other continuous performance indicators.

The model was estimated using robust (HC1) standard errors due to heteroskedasticity, confirmed by both the Breusch-Pagan and White tests.

Assumptions Checking

Heteroscedasticity

The Breusch–Pagan and LM (White) tests were conducted. In both cases, the p-value was 0.0000, indicating the presence of heteroscedasticity. To address this issue, heteroskedasticity-robust standard errors (HC1) were applied.

Multicollinearity

The VIF test showed that all values were below 2.5, which is well under the common thresholds of 5 and 10. Therefore, there is no multicollinearity, and all variables were retained in the model.

Misspecification

The Ramsey RESET test (with squared, cubic, and combined terms) produced significant p-values below 0.05, suggesting that the model is misspecified. This may indicate omitted variables or the need for a transformation (e.g., logarithmic). A log transformation of the dependent variable was therefore applied.

Normality of residuals

According to histograms, the Q-Q plot, and formal tests (Jarque–Bera, Shapiro–Wilk, etc.), the residuals are not normally distributed ($p \approx 0$). After applying the log transformation, the distribution improved but did not achieve full normality. Given the large sample size, this violation is not considered critical.

Model Interpretation

The following log-linear model was used:

$\ln(G+A+1)_i = \beta_0 + \beta_1 \cdot Age\ i + \beta_2 \cdot Min_i + \beta_3 \cdot Prg_p_i + \beta_4 \cdot Prg_r_i + \gamma_1 \cdot DPos_{1i} + \gamma_2 \cdot DPos_{3i} + \gamma_3 \cdot DPos_{4i} + \delta_1 \cdot DComp_{1i} + \delta_2 \cdot DComp_{2i} + \delta_3 \cdot DComp_{3i} + \delta_4 \cdot DComp_{4i} + \varepsilon_i$;

Where:

- "$\ln(G+A+1)_i$" is the natural logarithm of the number of goals and assists for player i;
- "Age_i" is the age of the player;
- "Min_i" is the total number of minutes played;
- "Prg_p_i" is the number of progressive passes;
- "Prg_r_i" is the number of progressive runs;
- "$DPos_{1i}$", "$DPos_{3i}$", "$DPos_{4i}$" are dummy variables for position (Forward, Defender, Goalkeeper);
- From "$DComp_{1i}$" to "$DComp_{4i}$" are the dummy variables for league (Premier League, La Liga, Ligue 1, Serie A);
- "ε_i" is the error term.

Note: Midfielders ($DPos_2$) and Bundesliga ($DComp_5$) were excluded as baseline categories.

This model estimates how different player characteristics - such as age, playing time, actions on the ball, position, and league, all the above effect offensive productivity, measured as the log of goals_and_assists +1. Since it is a log-linear model, coefficients for variables can be interpreted as approximate percentage changes:

For small coefficients ($-0.15 < \beta < 0.15$): $\%\Delta \approx \beta \times 100\%$

For larger coefficients: $\%\Delta = (e^{\beta} - 1) \times 100\%$

Age Effect:

Coefficient: 0.000497747 (p-value = 0.8112)

Age does not have a statistically significant effect on the goal contribution, because its p-value is above 0.05 threshold. Each additional year of life is associated with an increase in the goal contribution by only 0.05%, but this effect does not statistically differ from zero.

Minutes Played (Min) Effect:

Coefficient: 0.000630896 (p-value = 4.57e-20)

A very significant positive effect. Each additional minute played increases the number of goals scored by approximately 0.063%. During 90 minutes of a match, the expected number of goals scored increases by about 5.7%.

Progressive Passes (prg_p) Effect:

Coefficient: 0.00198754 (p-value = 0.0801)

Progressive Passes do not have a statistically significant effect on the goal contribution, because its p-value is above 0.05 threshold. Each additional progressive pass increases the number of goals scored by about 0.20%, but this effect does not statistically differ from zero.

Progressive Runs (prg_r) Effect:

Coefficient: 0.0137425 (p-value < 0.001)

Highly significant effect. Each additional progressive run increases the goal contribution by about 1.38%. This suggests that progressive runs are about 7 times more valuable for goal contributing than progressive passes.

Forward (DPos_1) Effect:

Coefficient: 0.154111 (p-value < 0.001)

Forwards score about 16.7% more goals than midfielders (e^ 0.154111 - 1 = 0.1667), while other factors remain constant.

Defender (DPos_3) Effect:

Coefficient: -0.183734 (p-value < 0.001)

Defenders have a 16.8% lower goal percentage than midfielders (e^{-0.183734} - 1 = -0.1679).

Goalkeeper (DPos_4) Effect:

Coefficient: -0.329679 (p-value < 0.001)

Goalkeepers score about 28.1% fewer goals than midfielders (e^{-0.329679} - 1 = -0.2809).

Premier League (DComp_1) Effect:

Coefficient: -0.0675697 (p-value = 0.0177)

The number of goals scored by Premier League players is about 6.5% lower than that of Bundesliga players.

La Liga (DComp_2) Effect:

Coefficient: -0.0758223 (p-value = 0.0067)

La Liga players have about 7.3% fewer goals scored than Bundesliga players.

Ligue 1 (DComp_3) Effect:

Coefficient: -0.0420923 (p-value = 0.1417)

There are no statistically significant differences from the Bundesliga players.

Serie A (DComp_4) Effect:

Coefficient: -0.0495793 (p-value = 0.0776)

There are no statistically significant differences from the Bundesliga players.

This model estimates how different player characteristics - such as age, playing time, actions on the ball, position, and league, all the above effect offensive productivity, measured as the log of goals_and_assists +1.

The regression was estimated using robust standard errors to account for heteroskedasticity.

Due to the large sample size (n = 2,176), many variables were statistically significant. Among the variables, minutes played and progressive runs had strong, positive, and statistically significant effects. This shows that players who are on the pitch longer and who carry the ball forward more often are more likely to contribute to goals and assists. These results are consistent with expectations in football, where active involvement and forward movement are key to offensive performance.

In contrast, progressive passes had a p-value of 0.0801, which is above the typical 0.05 significance level. This suggests that, holding everything else constant, making more progressive passes doesn't have a clear or consistent effect on a player's goal or assist output. This might be because such passes rely on team interaction and finishing by others, whereas runs are more individual.

Age also did not have a significant effect (p-value = 0.8112). Although we might expect younger or older players to perform differently, the results show no strong relationship once other factors like playing time and role are considered.

In terms of player position, all included dummy variables were significant. Compared to midfielders (which is the excluded category), forwards scored significantly more, while defenders and especially goalkeepers scored less. These effects match the general structure of football roles - attackers are more directly involved in scoring, and goalkeepers are not expected to contribute to offensive stats.

These regression findings allow us to reflect directly on the study's main hypothesis: that in leagues with higher defensive intensity, progressive runs (prg_r) would better explain offensive productivity (G+A) than progressive passes (prg_p).

In summary, while the data support the core idea that progressive runs are more impactful than progressive passes, they do not confirm that this advantage is greater in more defensively intense leagues. This suggests that, regardless of tactical context, ball-carrying actions play a key role in offensive effectiveness, reinforcing the value of individual initiative in modern football.

Model Fit and R^2

The adjusted R-squared of the model is approximately 0.38, meaning that 38% of the variation in the dependent variable is explained by the included predictors. This is a solid result for applied economic research, where outcomes are influenced by many unobservable factors. It suggests the model captures a meaningful part of the variation and provides useful explanatory power.

Conclusions

At the beginning, our hypothesis was that in leagues with higher defensive intensity, progressive runs would better predict player productivity (Goals and assists) than progressive passes. The analysis partially confirmed this idea. The coefficient for progressive runs was statistically significant, while progressive passes were not. However, the interaction between defensive intensity and either of these variables was not significant, since the dataset included the top 5 leagues in the world, and the average defensive intensity was approximately the same. This means that the number of tackles and interceptions does not affect the effectiveness of progressive passes in explaining player productivity in the strongest leagues, and progressive runs have a slightly stronger influence.

It turns out that the results suggest that, overall, progressive runs have a greater impact on player productivity than progressive passes, regardless of the league's playing style. One explanation could be that progressive runs reflect individual actions that a player can carry out without relying on teammates. This increases the chances that they will result in a goal or assist. Progressive passes, on the other hand, require successful interaction with another player and depend more on team dynamics. Additionally, the dataset may be overly representative of leagues like the Bundesliga, where quick transitions and direct attacks are more common, which further amplifies the role of progressive runs. In short, in the world's top leagues, in addition to the importance of team play, individual player performance significantly influences the outcome.

It's important to mention that this study has a number of limitations. The model does not include important variables such as team tactics or expected goals, which could better explain player productivity because expected goals would give a more complete picture of the danger and impact of a player's actions. Moreover, the data contains imperfections, including players with injuries, those who retired mid-season, or those with limited playing time. The use of OLS for a dependent variable that is right-skewed and count-based might also be suboptimal, even though a log transformation was applied. A different model such as exponential regression might have been more suitable.

Therefore, we suggest testing alternative model specifications in future research. Adding additional variables – such as team attacking style, possession share, and passing networks – could significantly improve the model's explanatory power. It would also be useful to shift focus to smaller, niche leagues and analyze regional differences in outcomes. A deeper league- or position-specific analysis could also help uncover how ball progression impacts productivity in different tactical environments. The study turned out to be fairly detailed and demonstrates explanatory potential.

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Note: "Clarifications of the proper interpretations of the normality tests, kernel density plots, Q-Q plots, and regression tables of the performed work in Gretl were supported by OpenAI's ChatGPT (GPT-4, custom econometrics setup). All AI-generated content was carefully reviewed, edited and validated by the authors"

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